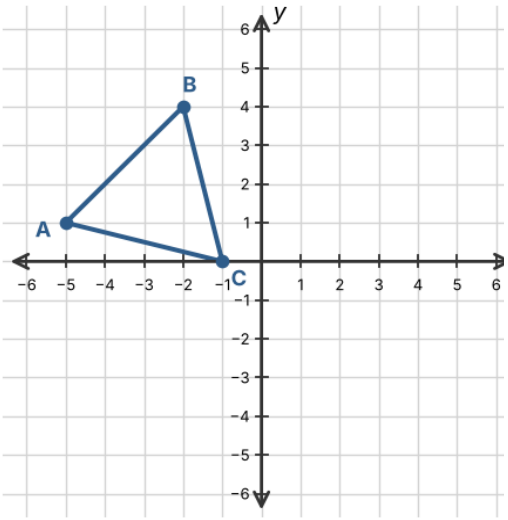
SC Honors Geometry

Chapter 4 Transformations Lab

Names: _____

You need: this sheet and a pencil. Everything you find today, you will use as a coordinate rule for the rest of the chapter.

Part 1 — Is a translation really a rigid motion?



1. Slide each vertex 6 units right and 3 units down. Plot and label the images. A′_____ B′_____ C′_____

2. Write the rule: (x, y) → (_____ , _____)

3. Distance formula: AB = _____ A′B′ = _____

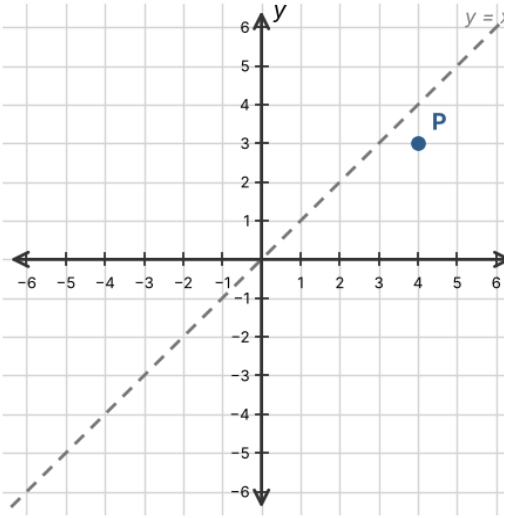
4. Slopes: AB ____ A′B′ ____ BC ____ B′C′ ____

5. Draw arrows AA′, BB′, CC′. Every arrow goes ___ right and ___ down, so every arrow has length _____.

- What stays the same: lengths, directions (slopes), or both? How do you know the angles also stay the same?

Rigid motion: a transformation that keeps every length and every angle the same. The image is congruent to the preimage.

Part 2 — Where the reflection rules come from



Fold the sheet on the line and press hard so P shows through, or count: how far is P from the line? Go the same distance past it.

1. Reflect P(4, 3) in the x-axis: P′_____ in the y-axis: P′_____

2. Fold on the dashed line y = x. Reflect P: P′_____

3. For the y = x image, find the midpoint of PP′: _____. Is it on y = x? ____

4. Slope of PP′: ____. Slope of y = x: 1. So PP′ is _____ to the line.

- The line of reflection is the _____ _____ of every segment joining a point to its image.
- Rules: x-axis (a, b) → _____ y-axis (a, b) → _____ y = x (a, b) → _____

★ **Think deeper:** Reflect P(4, 3) in the line x = 1, then reflect that image in the line x = 5. Where does P end up? What single transformation from today does the same thing? How far did P move compared to the distance between the two lines?

SC Honors Geometry

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Names: _____

Part 3 — Build the rule table

Use your rules from Part 2 and from Rotation Discovery. The last row IS the rule.

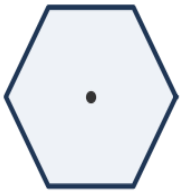
Point	in the x-axis	in the y-axis	in y = x	90° about O	180° about O	x-axis, then y-axis
(3, 5)						
(−2, 4)						
(6, −1)						
(a, b)						

▸ Two columns give the SAME image for every point. Which two? Use the (a, b) row to explain why.

▸ Which of these transformations are rigid motions? How can you tell without measuring?

Part 4 — Symmetry hunt (no scissors)

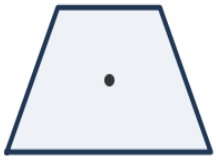
regular hexagon



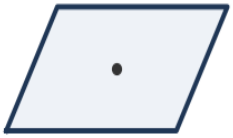
rectangle (not a square)



isosceles trapezoid



parallelogram



1. Draw every line of symmetry on each shape (dashed). A line works if folding on it makes the halves match.

2. Rotational symmetry: picture a pin at the dot. What is the smallest turn (less than 360°) that lands the shape back on itself?

	hexagon	rectangle	isos. trapezoid	parallelogram
lines of symmetry				
smallest turn				

▸ Name a figure from this page with rotational symmetry but NO line symmetry.

★ Think deeper: A regular polygon has n sides. How many lines of symmetry does it have? What is its smallest rotation? Check your answers with the hexagon.